Grade 9 Technology – Study Notes

These notes follow the standard Grade 9 CAPS content. Check them against your school's exam guidelines.

The design process and graphic communication

The design process (IDMEC)

Step	What you do
I – Investigate	Understand the problem: research, existing products, interviews, gather information
D – Design	Write a **design brief**, list **specifications** and **constraints**, draw several **ideas** (sketches) and choose the best
M – Make	Plan the steps, tools and materials (**flow chart / planning sheet**), then build the model or product safely
E – Evaluate	Test the product against the specifications; suggest improvements
C – Communicate	Present the solution with drawings, a portfolio and reports

- **Design brief** = a short statement of the problem and what must be designed. *Example: "I will design and make a model of a hydraulic lift that can raise a 200 g load 10 cm."*
- **Specifications** = requirements the product **must** meet (size, function, materials, cost, appearance, safety).
- **Constraints** = limitations (time, money, materials, tools available).
- **Evaluation** checks: does it work? Does it meet the specifications? Is it safe, attractive, affordable, environmentally friendly?

Graphic communication: drawing types

Drawing	Description	Used for
Freehand sketch	Quick drawing without instruments	Initial ideas
Oblique drawing	Front view drawn true shape; depth lines at **45°**, drawn at **half length** (cabinet oblique)	Simple 3D views
Isometric drawing	3D view; vertical lines stay vertical; horizontal edges at **30°**; all lengths true size	Showing an object in 3D
Single-point perspective	Lines meet at **one vanishing point** on the horizon	Realistic views (e.g. a road, a room)
Two-point perspective	Two vanishing points	Realistic views of buildings
Orthographic projection	2D views of an object: **front view, top view (plan), side view**; **first-angle projection** in SA	Working drawings for manufacturing
Exploded view	Parts drawn separated along their axes	Showing how parts fit together

Drawing conventions

Line type	Use
Thick (dark) continuous line	Visible outlines and edges
Thin continuous line	Dimension lines, projection lines, construction lines
Dashed (hidden) line	Hidden edges (not visible from that view)
Chain line (long dash, short dash)	Centre lines (circles, symmetry)

- **Dimensions** are in **millimetres (mm)**; units are not written on each dimension. Dimension lines have arrowheads touching the projection lines; numbers are written above the line.
- **Scale:** 1:1 = full size, 1:2 = half size (reduction), 2:1 = twice the size (enlargement).
- First-angle orthographic: the **top view is placed below** the front view, and the **left view on the right** of the front view.
- Use a **title block**: name, title, scale, date, drawing number.

Safety in the workshop

Wear safety goggles and closed shoes, tie back long hair, use tools for their purpose, cut away from your body, keep the work area clean, report injuries, never use broken tools or electrical equipment with damaged cords.

Mechanical systems and control

A **mechanism** changes an input motion or force into a useful output motion or force. A **system** = **input → process → output** (with feedback/control).

Types of motion: rotary (round and round — a wheel), **linear** (straight line — a train), **reciprocating** (back and forth in a line — a saw, piston), **oscillating** (swinging in an arc — a pendulum, windscreen wiper).

Levers

A lever is a rigid bar that turns around a **fulcrum** (pivot). It has an **effort** (input force) and a **load** (output force).

Class	Arrangement (middle item)	Examples
1st class	**Fulcrum** in the middle (E – F – L)	Seesaw, scissors, crowbar, pliers
2nd class	**Load** in the middle (F – L – E)	Wheelbarrow, bottle opener, nutcracker
3rd class	**Effort** in the middle (F – E – L)	Tweezers, fishing rod, braai tongs, your forearm

Memory aid **"FLE 123"**: the thing in the **middle** is F for class 1, L for class 2, E for class 3.

Mechanical advantage (MA) = load ÷ effort = effort arm ÷ load arm.

- MA > 1: the lever **makes work easier** (less effort needed). 2nd-class levers always have MA > 1.
- 3rd-class levers have MA < 1 — more effort, but greater movement and speed.
- **Moments:** load × load arm = effort × effort arm (when balanced).

Example: A wheelbarrow carries a 600 N load 0.4 m from the wheel (fulcrum). Effort is applied 1.2 m from the wheel. Effort = (600 × 0.4) ÷ 1.2 = **200 N**. MA = 600 ÷ 200 = **3**.

Gears

Gears are toothed wheels that transfer rotary motion. The gear turned by the input is the **driver**; the gear it turns is the **driven** gear.

- Two meshing gears turn in **opposite directions**. Add an **idler gear** between them to make the driver and driven gear turn in the **same** direction (the idler doesn't change the speed).
- **Gear ratio** = number of teeth on driven ÷ number of teeth on driver.
- **Small driver → big driven gear:** output turns **slower** but with **more force (torque)** — used when climbing hills on a bicycle.
- **Big driver → small driven gear:** output turns **faster** with less force.
- **Speed of driven gear** = speed of driver × (teeth on driver ÷ teeth on driven).

Example: A 20-tooth driver turns at 90 rpm and meshes with a 60-tooth driven gear. Gear ratio = 60 ÷ 20 = **3 : 1**. Speed of driven = 90 × (20 ÷ 60) = **30 rpm**, three times more torque.

Gear type	What it does	Example
Spur gears	Parallel shafts	Clocks, gearboxes
Bevel gears	Change direction of rotation by **90°**	Hand drill, egg beater
Worm and wheel	Large speed reduction; 90°; cannot be driven backwards	Guitar tuning pegs, gates
Rack and pinion	Rotary ↔ **linear** motion	Car steering, sliding gate

Pulleys and belts, sprockets and chains

- Pulleys joined by a belt turn in the **same direction** (crossed belt → opposite direction). Belts can slip (a safety feature).
- **Speed ratio** = diameter of driven pulley ÷ diameter of driver pulley. A small driver and a large driven pulley → slower, more force.
- **Sprockets and chain** (bicycle) do not slip.
- **Lifting pulleys:** a single fixed pulley changes the direction of force (MA = 1); a block and tackle with more pulleys gives MA > 1 (MA ≈ number of supporting ropes).

Other mechanisms

Mechanism	Motion change	Example
Cam and follower	Rotary → reciprocating	Car engine valves, toys
Crank and slider	Rotary ↔ reciprocating	Car engine piston, sewing machine
Linkages	Change direction or size of motion	Reverse-motion linkage, bell-crank linkage (90°), parallel-motion linkage (toolbox)
Ratchet and pawl	Allows rotation in one direction only	Winch, ratchet spanner
Screw	Rotary → linear; large force	Car jack, vice

Hydraulics and pneumatics

- **Hydraulic systems** use a **liquid** (oil or water); liquids **cannot be compressed**, so force is transferred efficiently. Examples: car brakes, hydraulic jack, excavator, dentist chair.

- **Pneumatic systems** use **air (gas)**; gas **can be compressed**, giving a softer action. Examples: air brakes on trucks, jackhammer, bus doors, tyre pump.
- **Pascal's law:** pressure applied to a liquid in a closed system is transmitted **equally in all directions**.
- **Pressure = force ÷ area** (P = F ÷ A), in pascal (Pa) or N/m².
- **Mechanical advantage** of a hydraulic system = area of output piston ÷ area of input piston (output force ÷ input force).

Example: Input piston area = 2 cm², output piston area = 20 cm², input force = 10 N.

Pressure = 10 ÷ 2 = 5 N/cm² (the same everywhere)
Output force = 5 × 20 = **100 N** (MA = 10)
The output piston moves only **1/10** of the distance the input piston moves — you gain force but lose distance.

Advantages of hydraulics: large force from a small effort, smooth precise control. **Disadvantages:** leaks, oil can be messy and flammable.

Electrical systems and control

An electric circuit needs an **energy source** (input), **conductors**, a **control** (switch) and an **output device** (bulb, buzzer, motor). See also the Physical Sciences notes on current, voltage and resistance.

Components

Component	Function
Cell / battery	Energy source (input)
Switch (push, toggle, slide, reed)	Controls the circuit (on/off)
Resistor	Limits current to protect components
Variable resistor (potentiometer)	Adjusts current (volume, dimmer)
LDR (light-dependent resistor)	Resistance **decreases in bright light**; used as a light sensor (street lights)
Thermistor	Resistance changes with **temperature**; used as a heat sensor (fire alarm, kettle)
Diode	Lets current flow in **one direction only**
LED (light-emitting diode)	Gives light; needs a resistor in series; uses little energy
Transistor	Electronic **switch** or **amplifier**; a small base current switches a larger current
Capacitor	Stores charge for a short time (timing circuits)
Relay	An electromagnetic switch: a small current switches a bigger circuit
Buzzer / bulb / motor	Output devices (sound, light, movement)
Fuse / circuit breaker	Safety: breaks the circuit if current is too high

Resistor colour code

The first two bands are digits, the third band is the **multiplier** (number of zeros), the fourth is the **tolerance**.

Colour	Digit	Multiplier
Black	0	× 1
Brown	1	× 10
Red	2	× 100
Orange	3	× 1 000
Yellow	4	× 10 000
Green	5	× 100 000
Blue	6	× 1 000 000
Violet	7	
Grey	8	
White	9	

Tolerance: **gold = ±5%**, **silver = ±10%**.

Memory aid: **B**lack **B**ears **R**oam **O**ver **Y**ellow **G**rass **B**ut **V**iolets **G**row **W**ild (0–9).

Example: brown, black, red, gold → 1, 0, × 100 = **1 000 Ω (1 kΩ) ±5%**. *Example:* yellow, violet, orange, silver → 4, 7, × 1 000 = **47 000 Ω (47 kΩ) ±10%**.

Ohm's law and power

- **V = I × R** (voltage in V, current in A, resistance in Ω).
- **Power P = V × I** (watts, W). *Example:* a 230 V kettle drawing 8 A uses 230 × 8 = **1 840 W**.
- Series: $R_t = R_1 + R_2$, current the same, voltage shared. Parallel: voltage the same, current shared, total resistance smaller.

Example: An LED needs 2 V and 20 mA (0.02 A) from a 9 V battery. The resistor must drop 9 − 2 = 7 V. R = V ÷ I = 7 ÷ 0.02 = **350 Ω**.

Logic gates

Logic gates make decisions using two states: **1 = on (high)** and **0 = off (low)**. They can be built from switches.

Gate	Output is 1 when…	Switch model
AND	**Both** inputs are 1	Two switches in **series**
OR	**At least one** input is 1	Two switches in **parallel**
NOT (inverter)	The input is 0 (output is the opposite)	
NAND	NOT AND — output is 0 only when both inputs are 1	
NOR	NOT OR — output is 1 only when both inputs are 0	

Truth tables

A	B	AND	OR	NAND	NOR
0	0	0	0	1	1

A	B	AND	OR	NAND	NOR
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	1	0	0

NOT: input 0 → output 1; input 1 → output 0.

Examples: a machine that starts only when both safety buttons are pressed = **AND**; a doorbell that rings from the front **or** back door = **OR**; a light that comes on when it is **not** light outside = **NOT** (with an LDR).

Electrical safety and the environment

- **Fuses** melt and **circuit breakers** trip when the current is too high, preventing fires.
- The **earth wire** (green and yellow) carries current safely to the ground if a fault occurs. The **earth leakage** unit cuts power if current leaks (e.g. through a person).
- Plug wiring (SA): **brown = live**, **blue = neutral**, **green/yellow = earth**.
- Dangers: overloading plugs, damaged cords, water near electricity, illegal connections.
- Dispose of batteries properly (toxic chemicals); save electricity; use renewable energy.